



EVRAG: Enhanced Video Retrieval-Augmented Generation for Democratized Social Computing

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Abstract. Large Vision–Language Models (LVLMs) stand to catalyze transformative advances in social computing by enabling deep multi-modal understanding, precise retrieval, and contextually informed generation. Existing LVLMs excel at understanding video content but incur prohibitive computational and memory costs due to dense, fixed-rate sampling. To capture fine-grained visual details, most LVLMs sample at 1–2 frames per second, yielding an unmanageable frame volume for long videos. Although various sampling optimizations have been proposed, finding the optimal balance between efficiency and accuracy remains elusive. We introduce EVRAG: Enhanced Video Retrieval-Augmented Generation, a training-free pipeline that leverages scene-change detection to select one representative frame per scene, thereby drastically reducing frame volume without degrading performance. To support robust information retrieval, EVRAG employs ChromaDB to construct an auxiliary index of supplementary frames, enabling sub-millisecond query response times and seamless interoperability with existing LVLM architectures. This optimized indexing yields a minimal resource footprint, permitting deployment on cost-constrained and low-energy platforms and thus broadening access to advanced video–language AI. By lowering infrastructural barriers, EVRAG democratizes video-based social computing and enables organizations of all scales to perform real-time multimedia analysis. Experimental results demonstrate that EVRAG achieves performance on par with leading models while effecting multi-order-of-magnitude reductions in both computational and memory overhead.

Keywords: AI · RAG · democratized access · affordable infrastructure · social computing

1 Introduction

Recent statistics reveal that 54 percent of U.S. adults rely on social media as a news source at least occasionally [1]. The proliferation of video-centric social

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networks has created novel challenges in digital communication where robust, affordable multimedia analysis becomes increasingly essential [2]. In the past few years, Large Vision–Language Models (LVLMs) have been highly beneficial for comprehending multimedia. These models have shown remarkable capabilities in handling tasks that require joint processing of visual and textual data including, but not limited to, video summarization and multimodal question-answering [3]. As effective as these LVLMs have been, they can be quite resource-intensive because of their fixed-rate frame sampling (typically one or two frames per second), which can generate an overwhelming number of frames for long videos [4, 5].

While several approaches have been proposed to improve sampling efficiency, the computational and memory burdens remain high [6, 7]. As video length increases, the volume of sampled frames grows rapidly and makes real-time analytics increasingly impractical [6]. Motivated by these challenges, several studies have explored sampling optimizations aimed at retaining critical visual content while reducing redundant information. Nevertheless, to date, no principled method has struck an effective balance between computational efficiency and competitive model performance [6, 7].

To address the computational and memory challenges inherent to current LVLM sampling strategies, we propose EVRAG as a training-free, efficient pipeline that significantly reduces the number of processed frames without sacrificing the model’s understanding of long video sequences. Instead of sampling frames at regular intervals, EVRAG utilizes open-source scene-change detection algorithms to identify key transitions in videos and produce one representative frame per scene. This adaptive strategy reduces the overall computational load and preserves key contextual information in each frame.

In addition to the optimized sampling technique, EVRAG integrates a retrieval mechanism using ChromaDB to construct a dynamic knowledge base. By indexing and querying supplementary frames, the system reinforces the initial retrieval with additional context, thereby enhancing the overall robustness and precision of video-language model responses. Collectively, this dual approach directly addresses the trade-off between reducing resource consumption and maintaining high performance on long video understanding benchmarks.

EVRAG relies on ChromaDB for its efficient vector-store capabilities. Its highly optimized similarity search executes sub-millisecond queries over millions of embeddings [8]. The built-in Approximate Nearest Neighbor (ANN) algorithm maintains high retrieval accuracy while substantially reducing computational overhead associated with semantic matching. A minimal Python API integrates directly with our pipeline, and metadata tagging preserves essential temporal and contextual relationships. Together, these features equip EVRAG’s retrieval-augmentation layer with the performance, efficiency, and flexibility required for large-scale videolanguage tasks [9].

In summary, our EVRAG framework offers the following key contributions:

Enhanced Efficiency: By adopting scene-change-based frame extraction, EVRAG slashes computational and memory overhead compared to uniform sampling. Consequently, end users experience lower latency, infrastructure expenditures are reduced, and video-language AI becomes more widely accessible.

Ease of Integration: EVRAG features a modular, plug-and-play architecture that integrates effortlessly into existing LVLMM workflows, thereby accelerating its deployment across a broad spectrum of real-world scenarios.

In the subsequent sections of this paper, we elaborate on the technical implementation of EVRAG, present a thorough evaluation against leading LVLMM benchmarks, and discuss the implications of our findings for future research in efficient video-language processing.

2 Related Works

Recent advances in LVLMMs have significantly pushed the frontier in video understanding and retrieval. State-of-the-art models showcase impressive performance across a range of tasks. However, they depend on fixed, high-frequency frame sampling to capture temporal dynamics. That sampling strategy, while effective at preserving motion and continuity, incurs substantial computational and memory costs when applied to long-duration videos [6].

2.1 Frame Sampling Techniques

To mitigate the resource-intensive nature of exhaustive frame sampling, several works have explored adaptive strategies. For instance, the Temporal Segment Networks framework introduced a method to divide videos into segments and sample representative frames from each segment and reduce redundancy while still capturing essential temporal dynamics [10]. In a related vein, Chen et al. [11] proposed AdaFrame, which adaptively selects frames to optimize both accuracy and efficiency in video recognition. Similarly, Yu et al. [12] proposed a hierarchical, multi-modal summarization framework that dynamically selects representative frames at both coarse (segment) and fine (shot) levels based on multiple pre-trained encoders and complex fusion modules. Zhu et al. [13] utilized Retrieval-Augmented Generation (RAG) pipeline to store video frames using large-scale vector indices in a multi-GPU environment. Although these methods have demonstrated effectiveness, they often require dedicated training or complex model adjustments, limiting their direct applicability within existing LVLMM pipelines.

In recent years, AI agents have been leveraged for dynamic frame sampling in video analysis. Jeoung et al. [14] introduced the Adaptive Video Understanding Agent (AVUA) to enhance both sampling efficiency and reasoning capabilities. However, AVUA requires reasoning queries to be specified before video processing begins which leads to an inflexible and impractical workflow. Moreover, AVUA and similar frameworks, such as MAF-ID [15], depend on substantial infrastructure resources and this constrains their accessibility and affordability in on-the-fly applications.

2.2 Scene-Change Detection in Video Analysis

Another promising direction has been the application of scene-change detection algorithms to identify key transition points in a video. Research proposed by Wang et al. [16] presented a comprehensive review on Shot Boundary Detection which underscores the importance of detecting semantic shifts to isolate contextually rich frames for summarization and analysis. Similarly, Fassold et al. [17] presented a unified, non-learning algorithm that simultaneously detects shot boundaries, determines sampling structure (progressive/interlaced/pulldown), and extracts dynamic keyframes at speeds four times faster than standard mechanisms. Open-source implementations of these detection methods have lowered the barrier to deployment, offering a training-free alternative that inherently aligns with the natural boundaries within video content. Such approaches not only reduce the number of frames processed but also ensure that the selected frames carry substantial contextual significance.

3 Model Description

We introduce EVRAG, a novel training-free pipeline designed to enhance the efficiency and performance of LVLMs. As illustrated in Fig. 1, EVRAG comprises two primary processing streams that operate in parallel: a speech-text stream and a visual stream.

In the speech-text stream, we first isolate the audio track from the input video and process it with Whisper, a state-of-the-art open-source automatic speech recognition model. The generated transcript, combined with the user’s query, constitutes the textual input to the LVLM, thereby enabling contextual reasoning over spoken content.

Simultaneously, the visual stream performs a fine-grained video segmentation by cutting clips and detecting scene changes with a scene-change detection algorithm. Instead of using dense and fixed-rate sampling, EVRAG employs a content-change detection algorithm to select representative frames that capture only meaningful visual transitions. This detector works by computing frame-to-frame differences in the HSV color space and flagging a scene cut when the average change in color and intensity exceeds a specified threshold. Specifically, we set a pixeldifference threshold and impose a minimum scene length of 15 frames so that, once a cut is detected, at least fifteen frames must pass before another cut can be registered. This reduces spurious detections caused by noise or brief camera motions. Once the start and end frames of each scene have been identified, we take the frame at the temporal midpoint of the scene as its representative and save these images in chronological order. These saved frames are then used as the visual input to the LVLM. In parallel, all extracted video clips are encoded using CLIP and stored in ChromaDB, forming a dedicated video-centric knowledge base.

To improve the quality and factual grounding of generated responses, we introduce a more efficient retrieval mechanism. After the LVLM produces an initial output, we construct a refinement prompt that encourages the model to

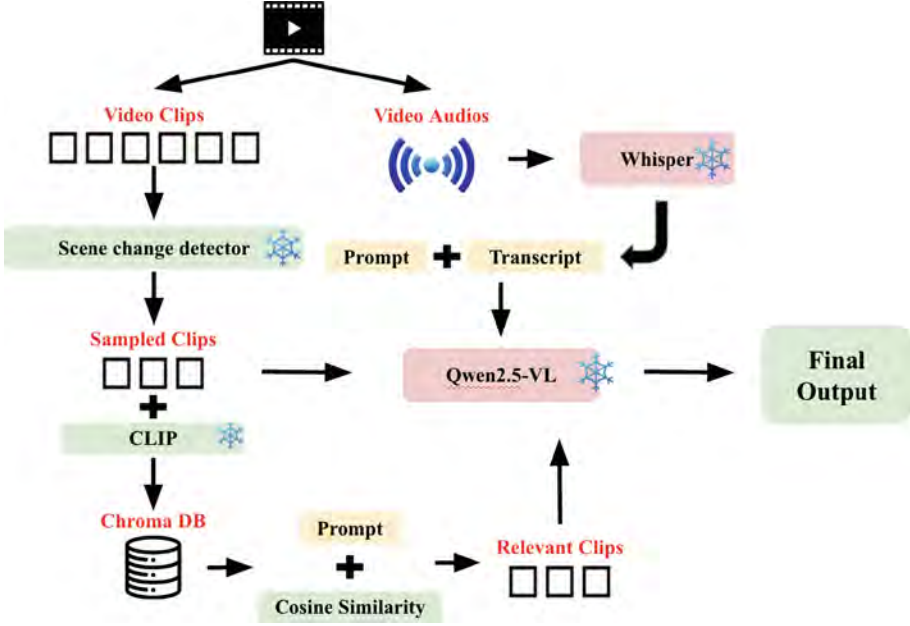


Fig. 1. Overview of the proposed EVRAG pipeline. The system leverages scene-change detection to extract representative frames and applies a retrieval mechanism powered by ChromaDB for robust, efficient video-language understanding

reconsider its answer in light of additional evidence. Specifically, we retrieve semantically relevant video clips from ChromaDB by computing cosine similarities between the original prompt and the CLIP embeddings of stored clips. These retrieved clips, along with the refinement prompt, are re-fed into the LVLM to generate a more accurate and contextually grounded final response.

EVRAG is model-agnostic, computationally efficient, and easy to integrate into existing LVLM frameworks, making it a practical solution for long-form video understanding with minimal overhead.

4 Experiments and Results

4.1 Video-MME Benchmark and Qwen2.5 Multimodal Foundation Model

In our paper, we compare EVRAG performance to Qwen2.5 LVLM using Video-MME benchmark. Video-MME is the first full-spectrum multimodal evaluation benchmark for Multimodal Large Language Models (MLLMs) in video analysis [18]. It covers aspects such as diversity in video types, duration in the temporal dimension, breadth of data modalities, and quality of video annotations [18]. Qwen2.5 constitutes a substantial advancement in multimodal foundation

models. It delivers marked improvements in both video retrieval and augmented generation tasks [19]. The architecture incorporates several key features:

- **Strong semantic video comprehension:** Qwen2.5 demonstrates remarkable temporal reasoning capabilities and it processes videos with high accuracy and good contextual understanding.
- **Efficient retrieval mechanisms:** Qwen2.5 utilizes dynamic-rate frame sampling for indexing videos. This technique, in combination with advanced search algorithms, reduces query latency while preserving retrieval precision.
- **Enhanced cross-modal integration:** The model unifies text, image, and video streams together. This enables more coherent and natural augmented outputs.
- **Scalable model:** Qwen2.5 supports deployment across various hardware configurations without sacrificing performance.

Several LVLm benchmarks demonstrate that Qwen2.5 outperforms existing models on standard video retrieval benchmarks. Qwen2.5 achieves outstanding results on VideoQA, temporal localization, and content-based video retrieval tasks. These results make it an ideal candidate for next-generation video retrieval augmented generation systems [19].

4.2 Experimental Results

We conduct a comprehensive evaluation of the EVRAG framework across a diverse set of video understanding tasks, with particular emphasis on long-form video content. Our primary goal is to assess both the computational efficiency and performance of the proposed approach in comparison to state-of-the-art LVLms, specifically Qwen2.5. We additionally incorporated the long-video accuracy statistics for GPT-4o as reported by the Video-MME benchmark [18].

4.3 Experimental Setup

For our experiments, we used a balanced set of videos comprising various genres and complexity levels. We categorized these videos into three groups based on duration: short (0–2 minutes), medium (2–5 minutes), and long (5+ minutes). Then, we evaluate model performance using standard accuracy metrics. It’s worth to note that all experiments were conducted on identical hardware configurations to ensure fair comparison.

4.4 Result Analysis and Discussion

Table 1 presents the comparative results between Qwen2.5 and our proposed EVRAG framework. The findings reveal several significant advantages of our approach:

Accuracy Improvements: EVRAG achieves 63.50% accuracy, a substantial 16.06%-point improvement over Qwen2.5’s 47.44%. When evaluated on long-videos from the Video-MME dataset, EVRAG achieved an accuracy of 80%,

Table 1. Performance comparison between Qwen2.5 and our proposed EVRAG framework. EVRAG achieves superior accuracy while drastically reducing inference time across videos of varying lengths.

Metric	Qwen2.5	EVRAG	Improvement
Accuracy (%)	47.44	63.50	+16.06
Inference Time (SECONDS)			
Short videos	63	5	12.6× faster
Medium videos	345	33	10.5× faster
Long videos	912	83	11.0× faster

substantially outperforming Qwen2.5 (53.33%) and GPT-4o (65.3%). This performance gap underscores EVRAG’s superior capability in handling extended video content. This enhancement shows that our scene-based adaptive sampling strategy not only reduces computational requirements but also facilitates more effective visual understanding. It does so by detecting semantically salient transitions in the video stream and selecting representative frames that each convey discrete, information-rich content. By focusing on these keyframes, EVRAG eliminates redundancy while retaining critical visual context.

Computational Efficiency: Perhaps EVRAG’s most notable advantage is its dramatic reduction in inference time. For short videos, processing time falls from 63 s to just 5 s (12.6× faster). Similar gains are observed for medium (345 s to 33 s, 10.5x faster) and long videos (912 s to 83 s, 11.0x faster). The greatest speedup in short videos suggests that our scene-detection algorithm is particularly efficient when scenes change rapidly or information density is high. Importantly, the relative efficiency gain remains stable as video duration increases, indicating that EVRAG scales well with longer content. This scalability is critical for real-world deployments that demand timely processing of lengthy video content.

The marked reduction in processing latency corresponds to a substantially diminished computational footprint and energy expenditure. This efficiency makes EVRAG eminently deployable on affordable, cost-sensitive hardware. In turn, such infrastructure affordability and performance democratize access to advanced videolanguage AI, enabling real-time analytics even in resource-constrained settings.

5 Limitations

Despite the promising performance of EVRAG, several limitations warrant acknowledgment:

Knowledge Base Scalability: As the number of stored video clips increases, the efficiency of ChromaDB retrieval may become a bottleneck. Future work should explore more scalable vector database solutions and optimized retrieval mechanisms for very large video collections.

Evaluation Scope: Our experiments encompass pre-recorded videos of varying lengths and complexities. Further evaluation across a wider array of video-understanding tasks and domains, such as live-streamed videos, would provide a more comprehensive assessment of EVRAG’s generalizability.

Hardware Dependencies: The absolute inference time improvements will vary based on hardware configurations. While the relative efficiency gains should remain consistent, the practical impact may differ across deployment environments with varying computational capabilities.

Despite these limitations, EVRAG constitutes a significant advancement in efficient videolanguage modeling, offering a compelling foundation for future inquiry. Its lightweight, training-free architecture not only enhances scalability but also makes the framework highly amenable to deployment in cost-sensitive environments.

6 Future Work

Building upon the promising results of EVRAG, several directions for future research emerge:

Dynamic Sampling Rate Adaptation: Although EVRAG presently relies on scene-change detection to guide frame selection, future work might develop more nuanced adaptive sampling strategies that modulate extraction frequency in accordance with scene complexity, motion dynamics, and semantic salience.

Multimodal Self-Refinement: EVRAG’s prevailing self-refinement methodology is predominantly restricted to visual modalities. Expanding this paradigm to encompass auditory signals, linguistic transcripts, and auxiliary multimodal inputs holds the potential to produce significantly more nuanced and semantically coherent refinements for multimedia content.

Domain-Specific Optimizations: Different video domains (e.g., news videos, sports, movies) may benefit from specialized sampling strategies tailored to their unique characteristics. Developing domain-specific variants of EVRAG could yield additional performance improvements for targeted applications.

7 Conclusion

We have presented EVRAG, a novel framework that reconciles the inherent tradeoff between computational efficiency and task performance in LVLMS. By harnessing scene-change detection to isolate representative frames and integrating ChromaDB-driven retrieval for supplementary context, EVRAG achieves substantial reductions in both processing time and memory footprint while simultaneously improving model accuracy.

Our empirical evaluation reveals that EVRAG surpasses the state-of-the-art Qwen2.5 model by 16.06% points in retrieval accuracy and accelerates inference by an order of magnitude across short, medium, and long video inputs. These results validate that adaptive, content-aware sampling effectively retains critical visual information and prunes extraneous frames that contribute minimally to task performance.

EVRAG’s utility extends well beyond quantitative performance improvements, offering distinct operational advantages for real-world deployment. Its

training-free design obviates the need for costly model retraining or invasive architectural modifications, facilitating frictionless integration into existing LVLM ecosystems. This minimal infrastructure requirement, together with near real-time performance, enables deployment on resource-constrained hardware and broadens access to advanced videolanguage AI.

In summary, EVRAG represents a significant step forward in democratizing social computing and enhancing multimedia analytics. By re-imagining how models sample and process videos, our framework paves the way for more accessible and practical video understanding systems that can be deployed at scale.

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